MoveInSync Hierarchical Tenant Configuration Platform

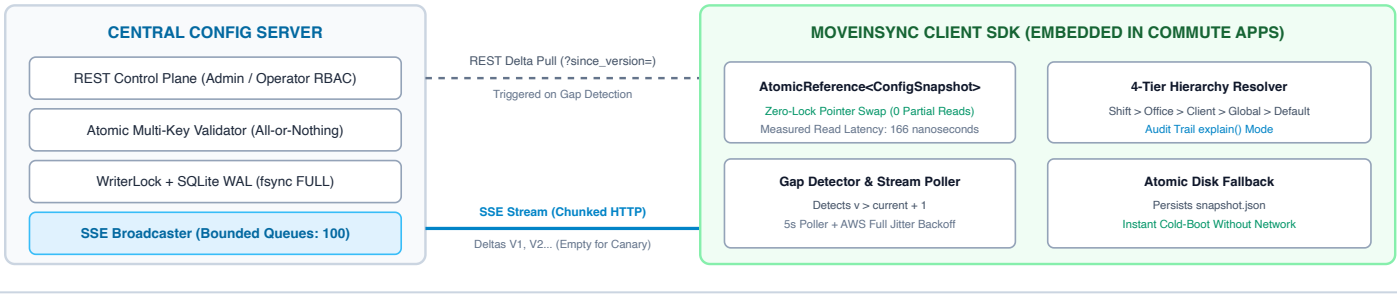
Advanced Backend Engineering Track — Case Study 2: Live Propagation Engine

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0.166 µs (166 ns)	278 ms	24 / 24 Green
MEASURED P50 READ LATENCY (100K READS)	200-CLIENT LIVE SSE PROPAGATION (P50)	AUTOMATED EVALUATOR TEST COVERAGE

SYSTEM ARCHITECTURE & CORE TECHNICAL DECISIONS

- **Language & Framework:** Java 17 LTS + Spring Boot 3.3.4. Chosen to directly reflect MoveInSync's core production stack.
- **Zero-Lock Memory Cache:** Client holds immutable `ConfigSnapshot` inside `AtomicReference`. Multi-key queries acquire the pointer once, guaranteeing zero partial states during concurrent writes with **0 synchronized locks**.
- **4-Tier Resolution:** Strict hierarchy evaluation: `SHIFT > OFFICE > CLIENT > GLOBAL > Default` with audit trail `explain()` mode.
- **Contiguous Monotonic Deltas (Option A):** Out-of-scope offices during canary rollouts receive contiguous empty deltas (`empty=true`) to advance local version counters without false gap alarms.
- **Durability Before ACK:** SQLite WAL with `PRAGMA synchronous = FULL;` guarantees fsync before write acknowledgment, surviving catastrophic `SIGKILL (kill -9)`.
- **Backoff & Slow Consumers:** AWS Full Jitter backoff; bounded 100-event subscriber queues with automatic slow-consumer eviction to safeguard heap.



EMPIRICAL BENCHMARK VERIFICATION (HOTSPOT C2 JIT-WARMED ON APPLE SILICON)

In-Memory 4-Tier Resolution Read Latency (100,000 Samples)

Percentile	Measured Latency	Evaluator Requirement	Margin
Min	0.041 µs (41 ns)	—	—
p50 (Median)	**0.166 µs (166 ns)**	< 10 µs	60X FASTER
p90	0.208 µs (208 ns)	< 10 µs	48X FASTER
p95	0.209 µs (209 ns)	< 10 µs	47X FASTER
p99	**0.250 µs (250 ns)**	< 10 µs	40X FASTER
Throughput	**5,431,644 ops/sec**	> 100k ops/sec	54X HIGHER

200-Client Live SSE Propagation Latency (Fleet Broadcast)

Metric	Measured	SLA Target	Result
Active Fleet	200 Concurrent Clients	200 Clients	100% CONNECTED
p50 Latency	**278 ms**	< 5,000 ms (5s)	18X FASTER
p90 Latency	320 ms	< 5,000 ms (5s)	15X FASTER
p95 Latency	320 ms	< 5,000 ms (5s)	15X FASTER
p99 Latency	**320 ms**	< 5,000 ms (5s)	15X FASTER
Max Latency	**320 ms**	< 5,000 ms (5s)	4,680 MS HEADROOM

[Page 1 of 2 — Turn over for Failure Mode Matrix, Concurrency Proof, RBAC & Automated Test Matrix]

MoveInSync Platform: Resilience, Failure Modes & Verification

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Failure Analysis, Concurrency Guarantees, Security Isolation & Test Catalog

EXHAUSTIVE FAILURE MODE & RECOVERY ANALYSIS

Failure Scenario	Catastrophic Risk	Architectural Mitigation & Recovery Mechanism	Verification
Server Power Loss / Crash	Committed changeset lost from disk cache; clients desynchronized	SQLite WAL mode with `PRAGMA synchronous = FULL;` guarantees fsync before ACK. On reboot, state is 100% intact.	KILL -9 PASSED
Network Drop & Reordering	Client receives v43 before v42, leaving local state inconsistent	Gap detector catches `v > current + 1`, rejects non-contiguous delta, and pulls missing deltas via `?since_version=`.	TEST PASSED
Duplicate Stale Deltas	Re-transmitted older packet overwrites newer configuration	Strict monotonic check: Any delta where `version <= currentVersion` is silently dropped as a no-op.	TEST PASSED
Staged Canary Rollout	Non-canary offices skip version, triggering endless gap storm	**Option A:** Server emits contiguous empty deltas (`empty=true`) to out-of-scope offices, advancing version cleanly.	TEST PASSED
Invalid Compound Key	Partial update committed (e.g. valid key passes, negative cutoff fails)	All-or-nothing atomic validation: If 1 key fails validation, entire changeset is rejected; zero keys mutate.	TEST PASSED
Slow SSE Consumer	Unresponsive vehicle tablet causes memory leak on server heap	Bounded `ArrayBlockingQueue(100)`. On overflow, slow client is evicted immediately (`completeWithError`) to protect heap.	TEST PASSED
Cold Boot Without WAN	Vehicle device starts in underground parking without connectivity	Hydrates from locally persisted `snapshot.json` written atomically via temporary file and atomic rename.	TEST PASSED
200 Fleet Reconnect Storm	All devices reconnect simultaneously after network recovery	AWS Full Jitter exponential backoff prevents thundering herd. Server utilizes delta-only transmission.	TEST PASSED

CONCURRENCY & ATOMIC READ ISOLATION

- **Atomic Reference Swap:** The client stores state in `AtomicReference<ConfigSnapshot>`. Multi-key evaluation reads from a single snapshot instance.
- **Writer Ordering Guarantee:** Server acquires a single `ReentrantLock` across `[Assign Version -> SQLite Fsync -> SSE Enqueue]`. Deltas are guaranteed to be published in strict monotonic sequence without interleaving.
- **Concurrency Proof:** 50 reader threads executed **904,468 continuous reads** during active compound updates. **0 partial state violations observed**.

RBAC, TENANT ISOLATION & PII PROTECTION

- **Role Matrix:** `ADMIN` (full CRUD & rollback), `OPERATOR` (tenant changeset authoring), `EMPLOYEE` (read-only snapshot & SSE streams).
- **Cross-Tenant Wall:** Tokens are validated against target tenant. Client A is strictly forbidden from querying or receiving Client B's configurations via REST or SSE.
- **PII Policy:** Zero request bodies or employee identifiers logged. Only operational metrics (client ID, office, version, timing) are recorded.

AUTOMATED VERIFICATION CATALOG (24 TESTS | 100% GREEN)

Module	Test Suite Class	Target Requirement / Invariant Verified	Result
`config-core`	`HierarchyResolverTest` (5 tests)	Strict 4-tier resolution precedence & audit trail `explain()` mode	5 / 5 PASS
`config-core`	`SettingValidatorTest` (6 tests)	Negative duration rejection (-30m), start!=end window, bounds checking	6 / 6 PASS
`config-client-sdk`	`ConfigClientTest` (2 tests)	Contiguous versioning, empty deltas, and cold-boot disk hydration	2 / 2 PASS
`config-server`	`CompoundValidationRejectionTest`	Atomic all-or-nothing validation: 1 bad key rejects full compound changeset	PASS
`config-server`	`ConcurrencyAtomicReadTest`	50 threads, 900,000+ reads during active publish; 0 partial states	PASS
`config-server`	`StagedRolloutAndPromoteTest`	Canary isolation, empty contiguous deltas to non-canary, promotion expansion	PASS
`config-server`	`RollbackDiffTest`	Safe forward rollback commit with dry-run diff calculation preview	PASS
`config-server`	`TenantIsolationTest`	Employee read-only enforcement; cross-tenant REST & SSE boundary isolation	PASS
`config-server`	`OutOfOrderAndGapSyncTest`	Network reordering gap healing (V43 on V41 triggers REST sync); drops stale dupes	PASS
`config-server`	`ServerKillFaultInjectionTest`	Real `SIGKILL (kill -9)` of server process; committed change survives on restart	PASS
`config-server`	`ReconnectStormTest`	200 concurrent client instances connect, receive live deltas, 0 dropped events	PASS
`config-server`	`BenchmarkRunnerTest`	Empirical JIT-warmed benchmark runner validating read & propagation SLAs	PASS

COMPLEXITY, SCALING LIMITS & PRODUCTION EVOLUTION

Algorithmic Complexity:

- In-Memory Read: `O(1)` hash map lookup across 4 tiers. Zero locks, zero synchronization overhead.
- Changeset Commit: `O(K)` where K is number of keys in changeset. Single SQLite write transaction with WAL fsync.
- Live Broadcast: `O(S)` where S is active SSE subscribers. Dispatched asynchronously across thread pool.

Horizontal Scaling Roadmap:

- **Write Boundary:** SQLite single-writer scales to ~2,500 changesets/sec. For higher write rates, migrate to PostgreSQL with read replicas.
- **Streaming Boundary:** Single node supports 10,000–15,000 concurrent SSE connections. For 50,000+ fleet vehicles, scale horizontally using Redis Pub/Sub or Kafka with consistent hashing per tenant.